

HAT-P-54b

R-1327

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_210225 · created 2026-07-31T20:09:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459270.7933 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.05
Transit depth (Rp/Rs)^2	1.97 +/- 1.39 [%]
Orbital Inclination (inc)	89.82 +/- 2.21
Airmass coefficient 1 (a1)	0.95 +/- 0.045
Airmass coefficient 2 (a2)	0.039 +/- 0.016
Scatter in the residuals of the lightcurve fit is	4.77 %
Best Comparison Star	#1 - [224, 79]
Optimal Aperture	2.39
Optimal Annulus	13.49
Transit Duration (day)	0.0696 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.05 vs 0.1572 (deviation 0.27 σ)
Duration fit vs published	0.0696 d vs 0.075 d (deviation 0.72 σ)
Mid-time O-C	4.9 min
Depth SNR	2.2
Residual scatter	4.77 %

Light curve

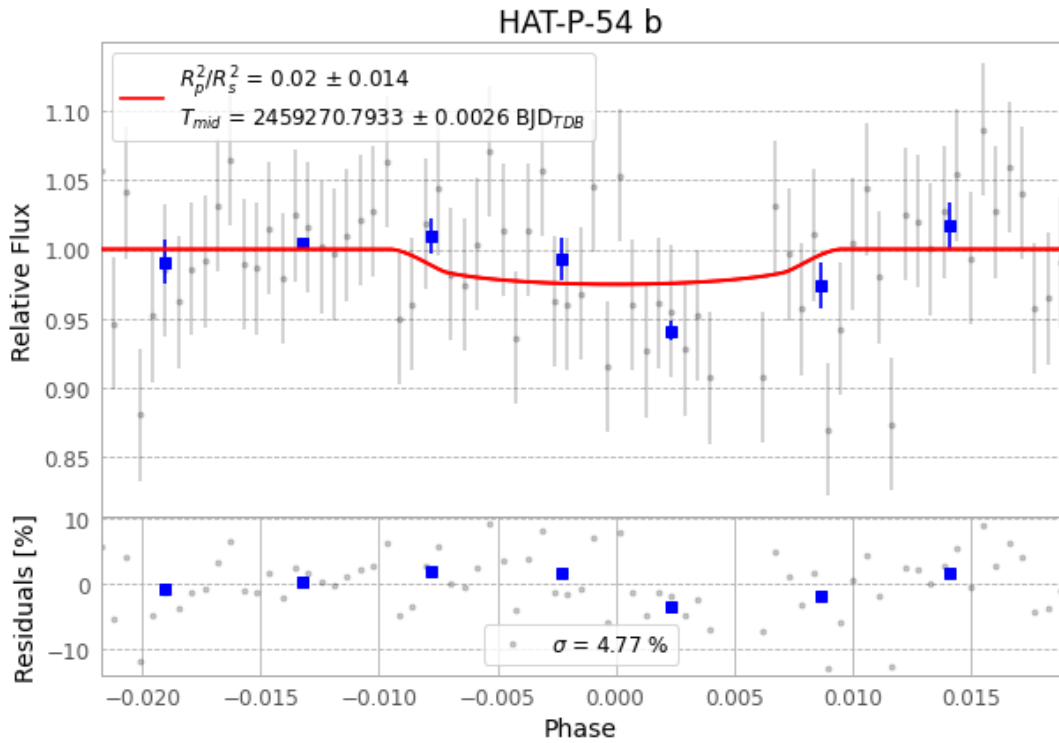


Figure 1 — FinalLightCurve_HAT-P-54 b_2021-02-24.png (SHA-256 c929551976aefc2d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7a08540225d7e412...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-54210225045712.FITS → HATP-54210225083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-210225.log (45f73ff9977d...), FinalLightCurve_HAT-P-54 b_2021-02-24.png (c929551976ae...), FinalLightCurve_HAT-P-54 b_2021-02-24.csv (47a8b7e62792...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:09Z.